Dialect Networks and Labor Market Sorting Among Internal Migrants in China

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Motivation

- **376 million** internal migrants (2020 Census)
- Hukou system creates formal local/migrant distinction
- **47%** of workers find jobs through friend/family referral
- Migrants face linguistic barriers *below* the national language
  - Mandarin (putonghua) = formal/official language
  - Local dialects (fangyan) = informal daily communication

Question: How does the *local* dialect affect migrant labor market sorting?

- 7–10 major dialect groups
  - Mandarin, Wu, Min, Cantonese, Hakka, Gan, Xiang
- Many **mutually unintelligible** in spoken form
- Mandarin = language of business, government, media
- Local dialect = daily life, informal networks

Concrete Example

A Sichuan migrant in Guangzhou:

- Speaks fluent Mandarin
- Cannot understand Cantonese
- Locked out of *local* informal networks

Two hiring channels coexist in Chinese cities:

Formal Channel

- Job boards, agencies, online postings
- Open to *all* workers
- Costly for firms $\rightarrow$ only productive firms invest
- Screens many candidates $\rightarrow$ hires the best

$\Rightarrow$ **Higher-paying firms**

Informal Channel

- Dialect networks, *laoxiang* referrals
- Only dialect speakers have access
- Free for firms $\rightarrow$ smaller firms rely on this
- Single referral $\rightarrow$ no screening

$\Rightarrow$ **Lower-paying firms**

1. Language and wages

- Mandarin: Gao and Smyth (2011); Xu and Liu (2023); dialect: Chen et al. (2014); Wei et al. (2019)
- $\rightarrow$ Nobody separates the two dimensions simultaneously

2. Referral networks and sorting

- Referrals yield *penalties* when formal channels efficient (Pellizzari, 2010; Lang and Yang, 2024); low-wage firms rely on informal hiring (Eliason et al., 2023)
- Network homophily $\rightarrow$ segregation (Buhai and van der Leij, 2023); co-group hiring in matched E-E data (Miller and Schmutte, 2021)

3. Between/within-firm wage decomposition

- AKM tradition (Abowd et al., 1999; Card et al., 2013; Song et al., 2019)
- $\rightarrow$ Never applied to a language variable

Data: Matched employer-employee survey, 13,700 workers, 875 firms, 15 cities, 3 waves (2013–2019)

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3. ⇒ **92% attenuation**: dialect operates entirely through firm sorting
4. Mandarin: positive in *both* (only 36% attenuation) — different channel

Contribution: Dialect = sorting variable; Mandarin = productivity variable

Conceptual Framework

Migrant arrives with education h, Mandarin proficiency m, dialect proficiency d

Mandarin as Productivity Input

- Lingua franca of formal business
- Improves communication with supervisors, clients, coworkers
- Generates premium at *any* firm
- $\Rightarrow$ Persists between **and** within firms

Dialect as Sorting Variable

- Determines access to local vs. hometown networks
- Channels migrants into specific firms
- Affects *which firm*, not earnings *within* the firm
- $\Rightarrow$ Entirely **between** firms

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4. **Migrant specificity:** Sorting effect for migrants; null for locals (placebo)
5. **Within-industry:** Sorting persists with industry FE (which firm, not which industry)

Explanation	Testable implication	Prediction
Employer discrimination	Within-firm pay gap	$\hat{\beta}_{\text{firm FE}} \neq 0$
Communication productivity	Within-firm pay gap	$\hat{\beta}_{\text{firm FE}} \neq 0$
Selection on unobservables	Oster $\delta < 1$	$\delta < 1$

Our results:

- $\hat{\beta}_{\text{firm FE}} \approx 0 \Rightarrow$ rules out discrimination and communication productivity
- Oster $\delta = 4.1 \Rightarrow$ unobservables need $4\times$ importance to explain away effect
- Mandarin shows *different* attenuation pattern $\Rightarrow$ not general ability

Motivation

Conceptual Framework

Data

Empirical Strategy

Sorting Evidence

Main Results

Additional Evidence

Robustness

Discussion and Conclusion

Data

- Repeated cross-sections: 2013, 2017, 2019
- Stratified sampling: city $\times$ industry $\times$ firm size
- **Employer questionnaire:** firm characteristics, industry, ownership
- **Employee questionnaire:** demographics, education, wages, job search, **dialect proficiency**, **Mandarin proficiency**
- Matched structure: multiple workers per firm $\longrightarrow$ within-firm estimation
- 15 cities across China (only Beijing, Chengdu, Guangzhou in all 3 waves)

Wave	Workers	Firms	Cities
2013	4,532	∼440	12
2017	4,558	∼300	7
2019	4,602	∼290	6
Total	**13,692**	**875**	**15**

- **46%** migrants (hukou registered in a different city)
- Wage regressions: 5,995 migrants in 810 firms
- Average firm size in sample: ∼10–15 workers

Dialect proficiency Self-reported proficiency in the *destination* dialect (1–5)

- 1 = cannot speak, 5 = fluent
- Measures linguistic distance, not general ability

Mandarin proficiency Same 1–5 scale; uniformly high (mean 4.63)

Friend/family referral Binary indicator; 47% of sample

Wages Log monthly earnings; mean $\approx$ 4,336 CNY

Controls Education (8 categories), age, age^2, gender, employee type

Key: Including Mandarin as control isolates the local-dialect-specific channel

	Migrants		Locals	
Dialect Level	*N*	%	*N*	%
1 (Cannot speak)	344	5.5	130	1.8
2 (Understand a little)	830	13.3	242	3.3
3 (Can speak some)	655	10.5	345	4.7
4 (Basic communication)	1,484	23.9	1,353	18.3
5 (Fluent)	2,910	46.8	5,330	72.0

- Migrants: **19%** low-dialect (levels 1–2); 47% fluent
- Locals: strong ceiling effect (**72%** fluent) $\longrightarrow$ less variation $\longrightarrow$ placebo

	Full Sample	Migrants	Locals
Age	32.8	30.1	35.0
Female	0.56	0.54	0.57
Education (1–8)	5.61	5.56	5.64
Dialect proficiency	4.27	**3.93**	**4.56**
Mandarin proficiency	4.63	**4.63**	**4.64**
Friend referral	0.47	0.44	0.49
Log monthly wage	8.22	8.24	8.19
N	13,692	6,240	7,412

Key asymmetry: Dialect differs sharply (3.93 vs. 4.56); Mandarin does not (4.63 vs. 4.64)

Empirical Strategy

Identification: Within-city variation in destination dialect proficiency among migrants

Key assumption: Conditional on city FE, Mandarin, education, demographics, variation in dialect reflects linguistic distance, not unobserved ability

Sequence of tests:

1. **Sorting evidence:** Within-firm homophily, permutation test
2. **Wage decomposition:** Between vs. within-firm
3. **Direct sorting test:** Dialect $\rightarrow$ firm wage premia
4. **Counterfactual:** Wage cost of sorting

HHI concentration:

$$\text{HHI}_{gct} = \sum_{f} s_{fgct}^2$$

Leave-one-out homophily:

$$\text{Homophily}_i = \frac{n_{g(i),f} - 1}{n_f - 1}$$

- Share of coworkers in same dialect group, *excluding own observation*
- Under random assignment: equals city-level population share ($\approx$19%)

Permutation test: Shuffle dialect labels within city-year cells, 5,000 draws

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Within-firm (Firm FE):

$$\ln w_{ifct} = \beta_1 \text{Dialect}_{ifct} + X'_{ifct}\gamma + \phi_f + \tau_t + \varepsilon_{ifct}$$

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$$\ln w_{ifct} = \beta_1 \, \text{Dialect}_{ifct} + X'_{ifct}\gamma + \phi_f + \tau_t + \varepsilon_{ifct}$$

- Same sample, same covariates — only difference is δ_c vs. ϕ_f
- **If sorting:** $\hat{\beta}_1$ significant in (1), vanishes in (2)
- Controls X: education, age, age^2, gender, Mandarin, employee type
- SE clustered at firm level

Selection on unobservables
- Mandarin controls for general language ability
- Oster $\delta = 4.1$ (unobservables need $4\times$ importance)

Hometown-specific effects
- Origin-province FE: coefficient *strengthens* ($-0.013 \rightarrow -0.018$)

Few clusters (15 cities)
- Wild-cluster bootstrap with Webb weights (Roodman et al., 2019)

Reverse causality
- Network sorting operates at hiring, before workplace exposure

Questions on framework and data?

Sorting Evidence

	Low Dialect (Levels 1–2)	High Dialect (Levels 4–5)	Difference
Mean HHI	0.032	0.031	−0.001
N	1,118	4,249	
t-statistic			−0.94
p-value			0.347

- No difference in how concentrated each group is across firms
- Low-dialect migrants are *not* in fewer firms
- Sorting is not about concentration — it's about **homophily**

	Low Dialect (Levels 1–2)	High Dialect (Levels 3–5)	Gap
LOO share of low-dialect coworkers	**44.4%**	12.8%	**31.6 pp**
N	1,108	4,803	
t-statistic			-31.9
p-value			<0.001

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- Under random assignment: $\approx$19% (sample share of low-dialect)
- Low-dialect workers: **44%** of coworkers are also low-dialect (2.3$\times$ random)
- Workers with similar dialect end up at the same firms

	Observed	Permutation Mean (SD)
LOO homophily gap	**0.315**	0.260 (0.007)
Standard deviations above		7.8
p-value		<0.0002

- Shuffle dialect labels within city-year cells, 5,000 times
- **0 out of 5,000** permutations exceed observed gap
- Observed gap lies 7.8 SD above permutation mean
- Rules out: firm size composition, city-level sorting, mechanical correlations

Main Results

	Migrants			Locals	
	(1) OLS	(2) City FE	(3) Firm FE	(4) City FE	(5) Firm FE
Dialect	-0.033^{***}	$\mathbf{-0.019}^{***}$	$\mathbf{-0.001}$	-0.013^{*}	-0.006
	(0.004)	(0.006)	(0.006)	(0.007)	(0.006)
Mandarin	0.055^{***}	0.025^{**}	0.016	0.038^{***}	0.024^{**}
	(0.010)	(0.012)	(0.011)	(0.010)	(0.009)
Education	0.073^{***}	0.067^{***}	0.031^{***}	0.087^{***}	0.054^{***}
	(0.004)	(0.007)	(0.006)	(0.008)	(0.006)
Female	-0.129^{***}	-0.139^{***}	-0.111^{***}	-0.143^{***}	-0.121^{***}
	(0.012)	(0.016)	(0.013)	(0.015)	(0.011)
N	5,995	5,995	5,934	7,250	7,198
R^2	0.41	0.47	0.67	0.52	0.69

SE clustered at firm level. Controls: age, age^2, employee type, year FE. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

	Between-Firm	Within-Firm
Dialect	$-\mathbf{0.0187}^{***}$	$-\mathbf{0.0014}$
	$(p = 0.003)$	$(p = 0.797)$

$$\text{Attenuation} = \frac{0.0187 - 0.0014}{0.0187} = \mathbf{92\%}$$

- Full 1-to-5 scale: **7.5%** between-firm gap vanishes entirely within firms
- Binary specification: low-dialect earns $+4.9\%^{***}$ between, $+1.0\%$ within
- Nearly the *entire* dialect–wage relationship = firm sorting

	Between-Firm	Within-Firm	Attenuation
Dialect	$\mathbf{-0.019}^{***}$	$\mathbf{-0.001}$	**92%**
Mandarin	$+0.025^{**}$	$+0.016$	**36%**

- Mandarin retains a positive coefficient within firms (36% attenuation)
- Consistent with direct **productivity** effect
- The two language variables behave **fundamentally differently**

Sharpest Falsification Test

If dialect proxied for general ability, *both* would attenuate similarly.

Instead: dialect $\rightarrow$ 92%, Mandarin $\rightarrow$ 36%.

Puzzle: Better dialect proficiency → *lower*-paying firms?

Interpretation: Dual labor market

- Dialect-proficient → local informal networks → smaller, local firms
- No local dialect → formal channels → larger, higher-paying employers

Consistent with:

- Wei et al. (2019): same negative sign in Guangdong manufacturing
- Chen et al. (2018): informal search carries wage penalty
- Long et al. (2017): network-based search constrains outcomes

Contrast: Referrals generally *improve* outcomes in developed countries (Burks et al., 2015). Chinese *laoxiang* networks may be informationally narrow.

	Between-Firm	Within-Firm
Migrants	$\mathbf{-0.019}^{***}$	$\mathbf{-0.001}$
Locals	-0.013^{*}	-0.006

- Local workers: weaker, marginally significant between-firm effect
- Much less variation (mean 4.56, 72% fluent)
- Dialect should not constrain local workers' firm sorting
- Supports **Prediction 4**: migrant specificity

Additional Evidence

DV: Leave-one-out firm mean log wage (average coworker wages)

	(1) Dialect Only	(2) Mandarin Only	(3) Both
Dialect	**−0.012*****		**−0.015*****
	(0.004)		(0.004)
Mandarin		0.013	0.020**
		(0.009)	(0.009)
N	5,993	5,993	5,993
R^2	0.582	0.582	0.583

City FE, controls. SE clustered at firm level. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Result: Dialect → **lower-paying** firms; Mandarin → **higher-paying** firms. Opposite signs.

	College		Non-College	
	City FE	Firm FE	City FE	Firm FE
Dialect	**−0.019*****	**−0.005**	−0.013	+0.009
	(0.007)	(0.007)	(0.011)	(0.009)
N	3,839	3,716	2,156	1,975

- **College migrants:** significant between-firm sorting ($p = 0.007$), null within
- **Non-college migrants:** insignificant in both specifications
- Sorting most consequential for workers whose credentials might otherwise grant access to better-paying firms

Within city $\times$ year $\times$ education cells: reassign low-dialect migrants to high-dialect firm distribution

	Mean Gap	Median Gap	N
All low-dialect migrants	-3.5%	-2.7%	1,084
Non-college	-2.7%	-1.8%	414
College	$\mathbf{-4.3\%}$	-3.9%	663

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Aggregate magnitude:

- Mean loss: **245 yuan/month** per affected worker (5.1% of mean wage)
- 290M internal migrants $\times$ 19% low-dialect $\approx$ 55M workers
- Back-of-envelope: $\sim$**160 billion yuan/year** nationally

Dialect $\rightarrow$ Friend referral:

- Without city FE: -1.3 pp per dialect point ($p = 0.012$)
- With city FE: -0.6 pp ($p = 0.36$) — attenuates, suggestive only
- Migrant-specific interaction: -1.6 pp ($p = 0.076$)

Dialect $\rightarrow$ Job-field match:

- $+0.043^{***}$ per dialect point ($p = 0.002$); constrains *which firm*, not occupational fit
- Locals: null ($\hat{\beta} = 0.013$, $p = 0.40$)

Caveat

Sign of β_1 is theoretically ambiguous: model operates through firm recruitment, not worker-reported search. Cross-sectional: cannot establish causality. Sorting is the central result.

Robustness

Test	Result	Table
Origin province FE	Strengthens ($-0.013 \rightarrow -0.018$)	A11
Wild-cluster bootstrap	$p_{\text{boot}} = 0.046$ (between)	–
Oster δ	4.1 (far exceeds 1)	–
Industry FE	Unchanged (-0.023)	A4
City $\times$ Year FE	Slight attenuation (-0.016)	A9
Binary dialect	$+4.9\%^{***}$ between, $+1.0\%$ within	A5
Year-by-year	Consistent negative all 3 waves	A6
Education heterogeneity	College-driven	A7

Bottom line: 92% attenuation is stable across specifications

	(1) City FE	(2) + Origin FE	(3) Firm FE	(4) + Origin FE
Dialect	-0.013^{*}	$\mathbf{-0.018}^{**}$	$+0.003$	$+0.001$
	(0.007)	(0.008)	(0.006)	(0.007)
N	4,099	4,099	4,038	4,032
R^2	0.52	0.52	0.72	0.72

- Subsample: 2013 + 2019 (hukou province observed)
- Absorbs all time-invariant province-of-origin confounders
- Coefficient **strengthens** from -0.013 to -0.018
- Province-of-origin effects *attenuate* the true sorting effect

Under $R_{\text{max}} = 1.3\tilde{R}^2$:

$$\delta = \mathbf{4.1} \quad \text{(wage regression)}$$

- Selection on unobservables would need to be **4.1 times** as important as the full control set to explain away the effect
- Full controls: education, Mandarin, age, gender, employee type, city FE
- Firm wage premium: $\delta = 2.6$ — also well above threshold
- Even under conservative $R_{\text{max}} = 1$: both $\delta > 1$

Standard threshold for concern: $\delta < 1$. We are far above.

Specification	Firm-Clustered p	Bootstrap p (city level, Webb weights)
Between-firm (dialect)	0.003	**0.046**
Within-firm (dialect)	0.797	**0.71**
Firm wage premium	0.001	**0.026**

- 15 cities = small number of clusters for city-level inference
- Webb weights designed for few-cluster settings (Roodman et al., 2019)
- Between-firm effect significant at 5% even with 15 clusters
- Within-firm effect remains firmly null

Industry FE:

	Dialect
City FE	-0.022^{***}
+ Industry FE	-0.023^{***}
Firm FE	-0.006

Industry FE does *not* attenuate $\Rightarrow$ sorting is **within** industries

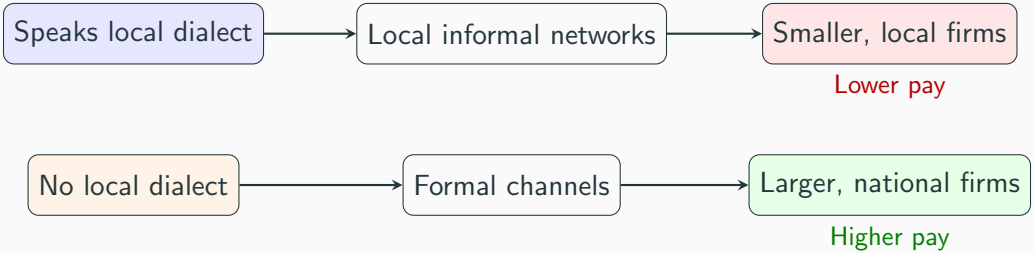
Year-by-Year:

Year	Dialect
2013	-0.008
2017	-0.022^{**}
2019	-0.018^{*}

Consistent negative in all waves (weaker in 2013: different cities)

Questions on results and robustness?

Discussion and Conclusion



- Contrast with developed countries: referrals generally *improve* outcomes
- Chinese *laoxiang* networks may be informationally narrow
- Dialect-based networks constrain rather than expand employer set

Two channels could generate dialect-based sorting:

Channel A: Regional
Speaks local dialect → local networks → locally rooted firms

Channel B: Enclave
No local dialect → *laoxiang* networks → co-provincial enclave firms

Test: Split firms by local share (fraction of workers from city's province; 695 firm-year cells)

- Firm premium: **−0.012** in high-local **and** **−0.012**** in low-local firms
- Interaction on firm premium: -0.002 ($p = 0.89$) — no difference
- Within-firm null holds in both subsamples

→ Sorting operates equally in both firm types

[Backup B13]

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All five consistent with: Dialect as a sorting variable, not productivity
Mandarin contrast: Provides the sharpest falsification

1. **Dialect training programs**
    - Could reduce segmentation by broadening network access
    - But current sorting coexists with reasonable occupational matching
2. **Formal job matching services**
    - Substitute for dialect-based informal networks
    - Digital platforms may already be weakening dialect dependence
3. **Mandarin education**
    - Direct within-firm productivity returns
    - Complementary to dialect/job-search interventions

Main finding: Dialect proficiency is associated with migrant wages entirely through between-firm sorting

- **92% attenuation** with firm FE
- Dialect = sorting variable; Mandarin = productivity variable
- 3–4% wage cost within city-year-education cells
- Robust to origin FE, wild bootstrap, Oster bounds, industry FE, firm composition

Limitations:

- Cross-sectional: cannot fully rule out selection on unobservables
- Cannot establish causal referral mechanism definitively

Future: Longitudinal data; role of digital job platforms

Thank you!

Comments welcome

Backup Slides

	Full Sample			Migrants			Locals		
	Mean	SD	*N*	Mean	SD	*N*	Mean	SD	*N*
Age	32.8	9.5	13,586	30.1	8.2	6,203	35.0	10.0	7,383
Female	0.56	0.50	13,650	0.54	0.50	6,238	0.57	0.50	7,412
Education	5.61	1.47	13,652	5.56	1.49	6,240	5.64	1.46	7,412
Dialect	4.27	1.11	13,623	3.93	1.27	6,223	4.56	0.86	7,400
Mandarin	4.63	0.67	13,641	4.63	0.67	6,231	4.64	0.67	7,410
Referral	0.47	0.50	13,633	0.44	0.50	6,236	0.49	0.50	7,397
Log wage	8.22	0.58	13,334	8.24	0.56	6,043	8.19	0.60	7,291
Match	3.32	0.97	10,361	3.34	0.99	4,854	3.31	0.95	5,507

	(1) Migrants	(2) Migrants	(3) Full Sample	(4) Locals
Dialect	-0.013^{**}	-0.006	0.002	0.004
	(0.005)	(0.007)	(0.008)	(0.008)
Dialect $\times$ Migrant			-0.016^{*}	
			(0.009)	
City FE	No	Yes	Yes	Yes
N	5,989	5,989	13,228	7,239
R^2	0.106	0.115	0.089	0.077

DV: indicator for friend/family referral. Controls: education, age, age^2, female, Mandarin, employee type, year FE. SE clustered at firm.

	Dialect → Match		Referral → Match	
	Migrants	Locals	Migrants	Locals
Dialect	0.043***	0.013		
	(0.014)	(0.016)		
Referral			−0.022	−0.148***
			(0.031)	(0.027)
N	4,671	5,381	4,671	5,381
R^2	0.105	0.084	0.103	0.089

DV: self-reported job-field match (1–5). City FE, controls. SE clustered at firm level.

Panel A: Manufacturing share by dialect group

	Low (1–2)	Medium (3)	High (4–5)
Migrants	43.6%	40.8%	36.6%
Locals	40.9%	42.8%	42.3%

Panel B: Regression (migrants, city FE)

	(1)	(2) + Referral
Dialect	-0.015^{*}	-0.015^{*}
	(0.008)	(0.008)
N	5,331	5,325

DV: manufacturing indicator. Marginal negative association ($p = 0.07$).

	(1) Separate FE	(2) City $\times$ Year FE
Dialect	-0.019^{***}	-0.016^{***}
	(0.006)	(0.006)
Mandarin	0.025^{**}	0.021^{*}
	(0.012)	(0.012)
N	5,995	5,995
R^2	0.470	0.475

Migrants only. Slight attenuation ($-0.019 \rightarrow -0.016$) but remains significant ($p = 0.010$). City-specific time trends do not explain the result.

	(1) City FE Between-Firm	(2) Firm FE Within-Firm
Low dialect (1–2)	0.049***	0.010
	(0.019)	(0.016)
Mandarin	0.018	0.016
	(0.012)	(0.010)
N	5,995	5,934

Positive coefficient: low-dialect migrants earn *more* between firms (consistent with negative continuous coefficient). Effect vanishes within firms.

	(1) 2013	(2) 2017	(3) 2019
Dialect	-0.008	-0.022^{**}	-0.018^{*}
	(0.009)	(0.011)	(0.010)
Mandarin	0.038^{**}	0.026	0.004
	(0.016)	(0.024)	(0.020)
Education	0.063^{***}	0.073^{***}	0.067^{***}
	(0.008)	(0.013)	(0.011)
N	1,949	1,894	2,152
R^2	0.359	0.352	0.319

Migrants only, city FE. Consistent negative direction across all waves.

Wave	Observations	Firms	Percent
2013	4,532	∼440	33.1%
2017	4,558	∼300	33.3%
2019	4,602	∼290	33.6%
Total	13,692	875	100.0%

- Balanced across waves (∼33% each)
- 15 cities total; only Beijing, Chengdu, Guangzhou in all 3 waves
- 46% migrants across pooled sample

City	Region	2013	2017	2019
Beijing	north	✓	✓	✓
Chengdu	southwest	✓	✓	✓
Guangzhou	south	✓	✓	✓
Shanghai	east		✓	✓
Zhengzhou	central	✓	✓	
Fuzhou	southeast	✓	✓	
Suzhou	east	✓		✓
Ji'nan	east	✓		
Xi'an / Xianyang	northwest	✓		
Taiyuan	north	✓		
Changchun	northeast	✓		
Qiqihar	northeast	✓		
Xiangyang	central	✓		
Shenyang	northeast		✓	
Dongguan	south			✓

Survey question: "How well can you speak the local dialect of this city?"

Level	Description
1	Cannot speak at all
2	Can understand a little
3	Can speak some
4	Can communicate at basic level
5	Fluent / native-like

- Self-reported: potential measurement error $\rightarrow$ attenuation bias
- 99.7% non-missing; available 2013, 2017, 2019 (not 2012, 2014, 2021)
- Same 1–5 scale for Mandarin proficiency

Setup: Compare R^2 progression as controls are added

	$\hat{\beta}$	R^2
OLS (no controls)	-0.033	0.41
+ City FE, controls	-0.019	0.47
+ Firm FE	-0.001	0.67

Under $R_{\text{max}} = 1.3 \times 0.47 = 0.61$:

$$\delta = \frac{\hat{\beta}_{\text{controlled}} \cdot (R_{\text{max}} - \tilde{R}^2)}{(\hat{\beta}_{\text{uncontrolled}} - \hat{\beta}_{\text{controlled}}) \cdot (\tilde{R}^2 - \dot{R}^2)} = 4.1$$

Interpretation: Unobservables need 4.1× the explanatory power of all observables to fully explain the dialect coefficient

	College		Non-College	
	City FE	Firm FE	City FE	Firm FE
Dialect	-0.019^{***}	-0.005	-0.013	$+0.009$
	(0.007)	(0.007)	(0.011)	(0.009)
Mandarin	0.038^{**}	0.024	0.020	-0.004
	(0.016)	(0.017)	(0.017)	(0.015)
Education	0.090^{***}	0.025^{*}	0.044^{***}	0.035^{***}
	(0.015)	(0.014)	(0.012)	(0.012)
Female	-0.109^{***}	-0.090^{***}	-0.192^{***}	-0.132^{***}
	(0.019)	(0.017)	(0.024)	(0.021)
N	3,839	3,716	2,156	1,975
R^2	0.475	0.670	0.432	0.700
Firms	693	570	558	377

Migrants only. SE clustered at firm level. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Panel A: Split by firm-year-cell median local share

	High local share			Low local share		
	OLS	Firm prem.	Firm FE	OLS	Firm prem.	Firm FE
Dialect	−0.017	−0.012	−0.005	−0.013	**−0.012****	+0.006
	(0.014)	(0.009)	(0.014)	(0.009)	(0.006)	(0.007)
N	1,428	1,428	1,390	2,613	2,612	2,608

Panel B: Interaction on firm wage premium (sorting margin)

Dialect	−0.012 (0.009)
Firm local share	0.050 (0.080)
Dialect × local share	−0.002 (0.014), $p = 0.89$
N	4,040

Migrants with origin data, firms with ≥3 workers. 695 firm-year cells. SE clustered at firm level.

Environment: Workers have Mandarin $m \sim G(m)$, dialect $d \in \{0, 1\}$, $m \perp d$. Firms have productivity $\varphi \sim F(\varphi)$. Output: $y = \varphi \cdot g(m)$, $g' > 0$. **Dialect does not enter production.**

Recruitment: Formal (cost $c > 0$, screens $n \geq 2$, hires best: $\bar{m}_F = \mathbb{E}[m_{(n:n)}]$) vs. informal (cost 0, single referral from $d=1$ pool: $\bar{m}_I = \mathbb{E}[m]$). Feasibility: $F(\varphi^*) \leq q$.

Threshold: $\varphi^* = c \,/\, [g(\bar{m}_F) - g(\bar{m}_I)]$. High-$\varphi$ firms use formal; low-φ use informal.

Predictions:

1. $\mathbb{E}[\varphi \mid d=1] < \mathbb{E}[\varphi \mid d=0] \rightarrow$ negative between-firm dialect effect
2. $w = \varphi \cdot g(m) \rightarrow$ zero within-firm dialect effect
3. $\text{Corr}(m, \varphi) > 0$ via formal selection $\rightarrow$ Mandarin persists with FE
4. Informal firms: all $d=1 \rightarrow$ excess dialect homophily (upper bound; test vs. random assignment)
5. Locals: $d=1$ always $\rightarrow$ no sorting (placebo)
6. φ^* industry-invariant $\rightarrow$ sorting persists within industry

Comparative static: $\partial\varphi^*/\partial c > 0$ — stronger sorting when formal channels expensive

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